

Gradient Descent Method

Inner Product: $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_{i=0}^n x_i \cdot y_i = x_1 \cdot y_1 + x_2 \cdot y_2 + \dots + x_n \cdot y_n$

Quadratic Function: $f(\mathbf{x}) = \frac{1}{2} \cdot \langle \mathbf{x}, \mathbf{Ax} \rangle - \langle \mathbf{x}, \mathbf{b} \rangle + c$

Gradient of Quadratic Function: $\nabla f(\mathbf{x}) = \frac{1}{2} \cdot \mathbf{A}^T \mathbf{x} + \frac{1}{2} \cdot \mathbf{Ax} - \mathbf{b}$

$\mathbf{A}$ is symmetric $\rightarrow a_{i,j} = a_{j,i}$

$\mathbf{A}$ is positive-definite $\rightarrow \langle \mathbf{x}, \mathbf{Ax} \rangle > 0, \mathbf{x} \neq \mathbf{0}$

$$\nabla f(\mathbf{x}) = \mathbf{Ax} - \mathbf{b}$$

f has a local minimum when

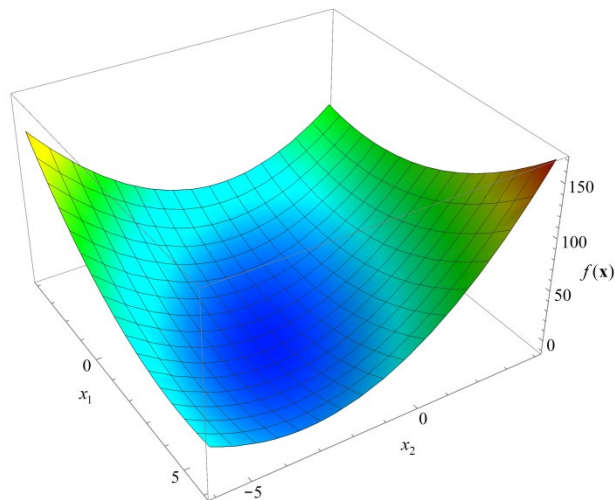
$$\mathbf{Ax} - \mathbf{b} = \mathbf{0}$$

$$\mathbf{Ax} = \mathbf{b}$$

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$$

$$\mathbf{b} = \begin{Bmatrix} 2 \\ -8 \end{Bmatrix}$$

$$c = 0$$



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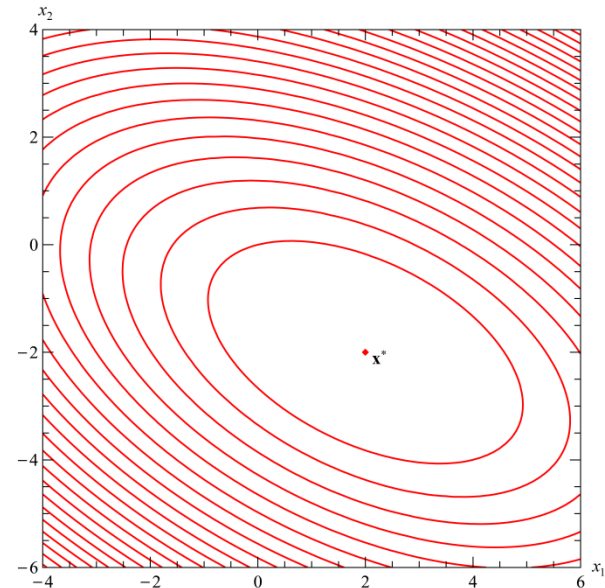
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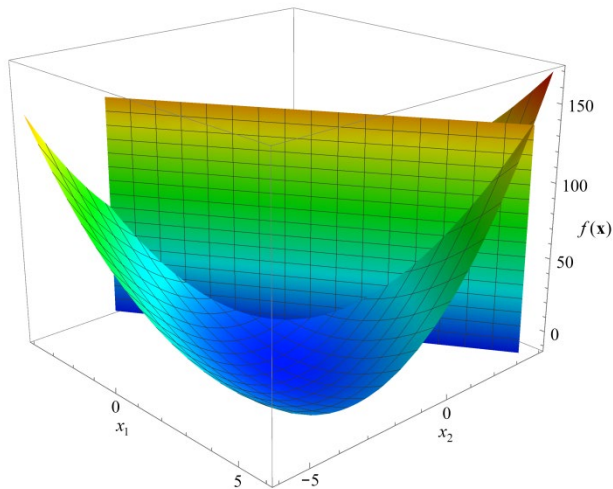
Gradient Descent Method

Recurrence Relation: $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha^{(k)} \cdot \mathbf{d}^{(k)}$, $\mathbf{d}$ is search direction.

Residual Vector: $\mathbf{r}^{(k)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(k)}$

Search Directions of Gradient Descent Method: $\mathbf{d}^{(k)} = \mathbf{r}^{(k-1)}$

Initial Guess: $\mathbf{x}^{(0)} = (-2, -2) \rightarrow \mathbf{r}^{(0)} = (12, 8) \rightarrow \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha^{(1)} \cdot \mathbf{r}^{(0)}$



$$\begin{aligned} \frac{d}{d\alpha^{(1)}} f(\mathbf{x}^{(1)}) &= \frac{\partial f(\mathbf{x}^{(1)})}{\partial x_1^{(1)}} \cdot \frac{\partial x_1^{(1)}}{\partial \alpha^{(1)}} + \frac{\partial f(\mathbf{x}^{(1)})}{\partial x_2^{(1)}} \cdot \frac{\partial x_2^{(1)}}{\partial \alpha^{(1)}} = \left\langle \nabla f(\mathbf{x}^{(1)}), \frac{d\mathbf{x}^{(1)}}{d\alpha^{(1)}} \right\rangle \\ &= \langle \nabla f(\mathbf{x}^{(1)}), \mathbf{r}^{(0)} \rangle = \langle \mathbf{A}\mathbf{x}^{(1)} - \mathbf{b}, \mathbf{r}^{(0)} \rangle \\ &= \langle \mathbf{A}(\mathbf{x}^{(0)} + \alpha^{(1)} \cdot \mathbf{r}^{(0)}) - \mathbf{b}, \mathbf{r}^{(0)} \rangle \\ &= \langle \mathbf{A}\mathbf{x}^{(0)} - \mathbf{b}, \mathbf{r}^{(0)} \rangle + \alpha^{(1)} \cdot \langle \mathbf{A}\mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle \\ &= \langle -\mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle + \alpha^{(1)} \cdot \langle \mathbf{A}\mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle \\ &= 0 \end{aligned}$$

$$\alpha^{(1)} = \frac{\langle \mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle}{\langle \mathbf{r}^{(0)}, \mathbf{A}\mathbf{r}^{(0)} \rangle}$$

Gradient Descent Method

$$\mathbf{A} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}, \mathbf{b} = \begin{Bmatrix} 2 \\ -8 \end{Bmatrix}, \mathbf{x}^{(0)} = \begin{Bmatrix} -2 \\ -2 \end{Bmatrix}$$

$k = 1$

$$\mathbf{r}^{(k-1)} = \mathbf{r}^{(0)} = \mathbf{b} - \mathbf{A}\mathbf{x}^{(0)} = (12, 8)$$

$$\alpha^{(k)} = \alpha^{(1)} = \frac{\langle \mathbf{r}^{(0)}, \mathbf{r}^{(0)} \rangle}{\langle \mathbf{r}^{(0)}, \mathbf{A}\mathbf{r}^{(0)} \rangle} = \frac{208}{1200} = 0.173333$$

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$$\begin{aligned} \mathbf{x}^{(k)} = \mathbf{x}^{(1)} &= \mathbf{x}^{(0)} + \alpha^{(1)} \cdot \mathbf{r}^{(0)} \\ &= (-2, -2) + 0.173333 \cdot (12, 8) \\ &= (0.08, -0.613333) \end{aligned}$$

$$\begin{aligned} \mathbf{r}^{(1)} &= \mathbf{b} - \mathbf{A}\mathbf{x}^{(1)} \\ &= (2.986666, -4.48) \end{aligned}$$

$$\varepsilon^{(1)} = \frac{\|\mathbf{r}^{(1)}\|_2}{\|\mathbf{r}^{(0)}\|_2} = 0.373333$$

